

Hybrid Multimodal Content Moderation and Integrity Verification Framework Using Deep Learning and NLP

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INTRODUCTION

Online platforms face rising threats like cyberbullying, misinformation, spam, harassment, and AI deepfakes. Traditional moderation methods fail to handle context and scale effectively. We are developing an AI-powered moderation system combining NLP and visual analysis to detect cyberbullying, misinformation, spam, harassment, pornography, and deepfakes in real time. Unlike traditional methods, it improves accuracy, reduces false positives, prevents harmful content spread, and enhances trust and safety online.

OBJECTIVES

This project focuses on the following primary objectives:

1. Detect harmful and manipulated content across text, images, and videos using multimodal AI.
2. Provide real-time content analysis and instant alerts for quick intervention.
3. Automate moderation with AI to reduce manual effort and improve efficiency.
4. Improve online safety, trust, and digital well-being by limiting harmful content.
5. Enhance performance through ensemble learning and a Chrome extension for real-time analysis.

METHODOLOGY

The methodology uses a hybrid multimodal AI framework combining NLP, deep learning, and ensemble learning for content moderation. Transformer-based models like BERT and RoBERTa analyze text, while EfficientNet, ViT, SwinV2, ConvNeXt, GRU, and Transformers detect manipulated images and deepfake videos through spatial, temporal, and contextual feature extraction. Ensemble fusion techniques improve accuracy, reduce false positives, enable real-time detection, and support integrated moderation across text, image, and video content.

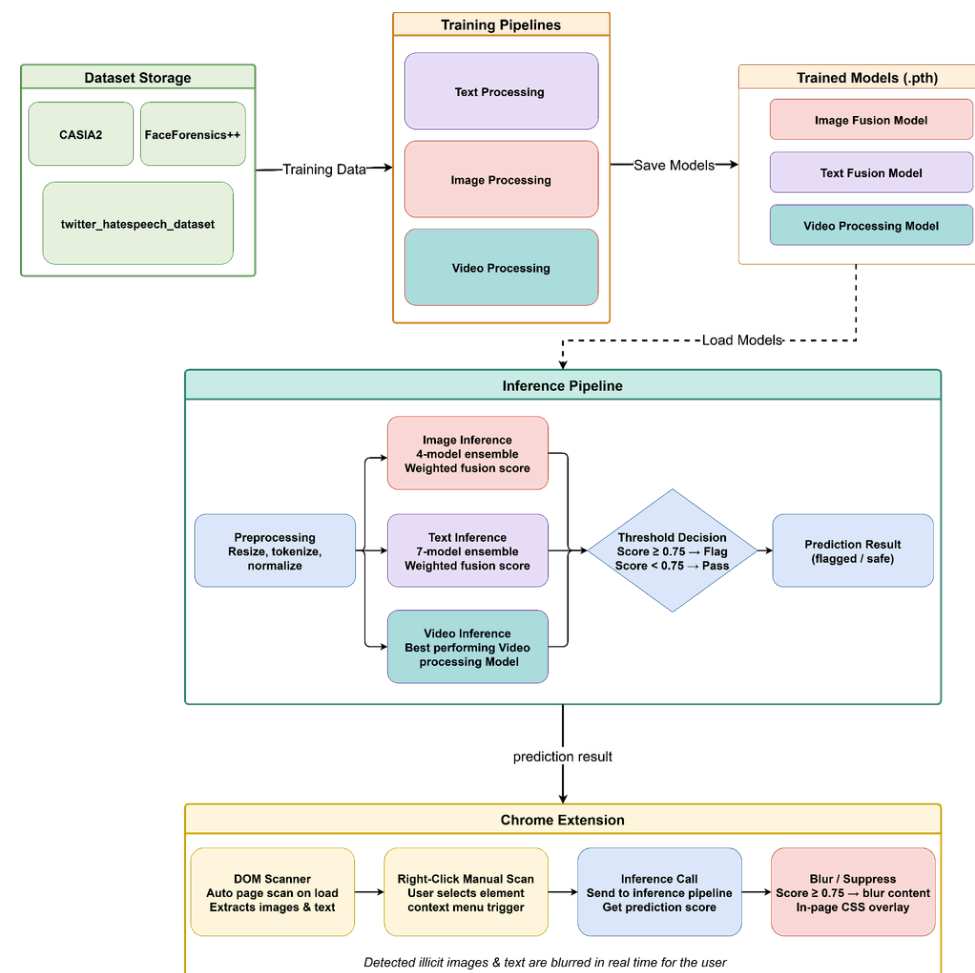


Fig. 1. Overall Architecture of the System

IMPLEMENTATION

The system was implemented using Python, PyTorch, TensorFlow, OpenCV, HuggingFace Transformers, and scikit-learn. Key modules include text classification, image forgery detection, deepfake video analysis, ensemble fusion, dataset preprocessing, real-time moderation, and a Chrome extension for live text and image scanning.

RESULTS

RoBERTa + BiLSTM with Attention Mechanism achieved 97% accuracy in hate speech detection. The EfficientNet + GRU + Transformer hybrid model achieved around 86.25% accuracy in deepfake video detection, while SwinV2, ConvNeXt, and EfficientNet-B3 performed best for morphed image detection. A Chrome extension enabled real-time text and image moderation during web browsing.

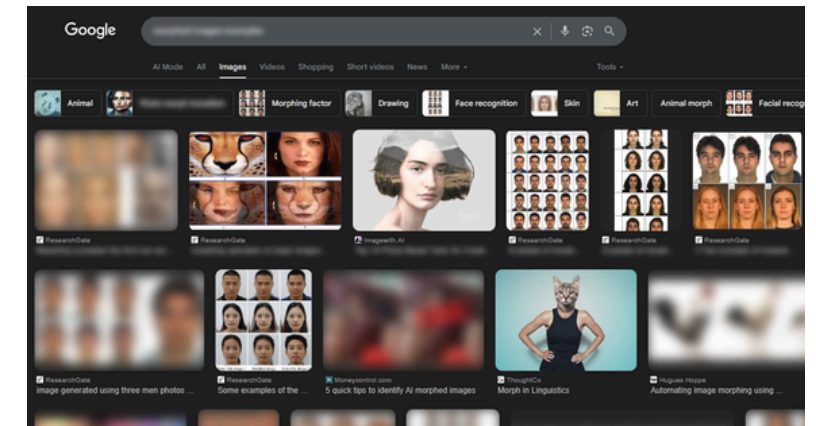


Fig. 2. Images that are Detected “Illicit” Blurred in Real-Time

CONCLUSION & FUTURE SCOPE

This research developed a multimodal AI system for detecting hate speech, deepfakes, and manipulated media using transformer and hybrid models. Future work focuses on improving efficiency, expanding datasets, enabling real-time deployment through browser extensions and APIs, and enhancing adaptability, scalability, and low-resource performance.

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